Pre-work · The Respiratory and Lymphatic Systems

The Respiratory System and The Lymphatic System. Read both Part A chapters first. The respiratory chapter carries over from the Module 3 packet, so treat it as a rebuild rather than a first read and give the new lymphatic material the longer half of your time. Both systems are pathway systems, so the order of the parts is the content.

Module 4 · 1 of 5 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the respiratory and lymphatic chapters into mini visuals. Eight chunks. Two chains carry the two pathways, and the pathways are where most of the marks are.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 GRID

Two ways of grouping the parts

Structure down the side, nose to alveolus in order. Upper or lower in one column, conducting or respiratory in the other. Circle every row where the two columns disagree, because those rows are the questions.

2 LADDER

The three regions of the pharynx

Three rungs, nasopharynx at the top. Beside each write what it carries, its epithelium, and its tonsil. Mark the rung where the epithelium changes and the rung where the air and food paths finally divide.

3 HUB

The larynx

The larynx at the hub. Spoke to the three single cartilages and the three paired ones, then to the vocal folds, the vestibular folds, and the glottis. Mark the one cartilage that is elastic and the one that is a complete ring.

4 CHAIN

The airway, trachea to alveolus

Trachea through terminal bronchiole, then on to the respiratory bronchiole, alveolar duct, and alveolus. Under the chain draw two bars, cartilage falling and smooth muscle rising, and mark the box where the conducting zone ends.

DRAWING 1

The lungs, the pleurae, and the bronchial tree

BRAIN DUMP FIRST

Two minutes. Both lungs with every lobe and fissure, the tree inside them, and the membranes around them.

Draw the trachea with its C-shaped cartilage rings and the carina below. Branch into both main bronchi and make the right one visibly wider and more vertical, then continue into lobar and segmental bronchi. Outline both lungs around the tree: three lobes and two fissures on the right, two lobes, one fissure, a cardiac notch, and the lingula on the left. Mark the apex, base, and hilum on each, then draw the parietal pleura against the chest wall, the visceral pleura on the lung surface, and the pleural cavity between them.

In your notebook, head the page **M4-1 The lungs, the pleurae, and the bronchial tree** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can say which lung an inhaled bead lands in, and point at the two features on your own drawing that decided it.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

After surgery for breast cancer a patient has had the axillary lymph nodes on one side removed. Months later that arm is persistently swollen. Name the condition and explain it from the structure of the system.

Answer. Lymphedema.

Reasoning. Start with what the lymphatic system does, not with the surgery. Lymphatic capillaries take up the fluid that leaks out of the blood capillaries and return it through collecting vessels, nodes, trunks, and ducts to the subclavian veins, so the whole thing is a one-way drain with no alternate route. Remove the axillary nodes and you remove the vessels that ran through them, and the lymph of that entire limb loses the road it was using. Fluid keeps arriving in the tissue spaces at the old rate but leaves more slowly, so it accumulates, and the swelling is the surplus. Ordinary edema would settle once its cause passed, which is what rules it out here, because the missing structure is not rebuilt. Notice that the arm swells and the trunk does not, and that tells you exactly where along the route the drainage was cut.

Set A · Name it

1. The scroll-shaped projections that swirl air in the nasal cavity are the ____________________.
2. The only complete cartilage ring in the airway is the ____________________.
3. The specialized lymphatic capillaries that absorb dietary fat in the small intestine are the ____________________.
4. The sac at the beginning of the thoracic duct is the ____________________.
5. Name the two primary lymphatic organs and say what matures in each.

5 SPLIT

Right lung against left lung

Lobes, fissures, size, and the feature unique to each side. Then hang the parietal pleura, the visceral pleura, and the pleural cavity off both branches, since those three are the same on both sides.

6 CHAIN

The pathway of lymph, tissue space to bloodstream

Six boxes with arrows, lymphatic capillary through to the subclavian vein. Mark the box where the lymph is filtered, and write beside the last box that this is the single point where a one-way system finally meets the blood.

7 HUB

A lymph node

One bean-shaped node at the hub. Spoke to the capsule, cortex, medulla, and hilum. Draw several afferent vessels entering through the capsule and one efferent vessel leaving at the hilum, then write the three regions where nodes cluster.

8 GRID

Primary against secondary lymphatic organs

Organ down the side, six rows. Group, location, and role across the top. Only two rows can say primary, and the reason they do is the answer to the whole grid.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

An alveolus and the respiratory membrane

BRAIN DUMP FIRST

One minute. Every cell of the alveolar wall and the three layers a gas crosses.

Draw one alveolar sac with several alveoli opening into it, at large scale. Line the wall with flat type I cells, add a few cuboidal type II cells, and put an alveolar macrophage on the surface. Wrap a pulmonary capillary against the outside, then enlarge the point where the two touch and label the alveolar epithelium, the fused basement membranes, and the capillary endothelium.

In your notebook, head the page **M4-2 An alveolus and the respiratory membrane** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at what emphysema destroys on your own drawing and say why the loss is permanent.

Set B · Predict it

1. The chest wall is punctured and air enters the pleural cavity. Predict what happens to the lung and name the condition.
2. A patient loses the use of the skeletal muscles of one leg. Predict the effect on lymph return from that limb and say why.
3. A premature infant lacks surfactant. Predict what happens to the alveoli between breaths and name the cell that should have made it.
4. A blow to the lower left ribs is followed by heavy internal bleeding. Name the organ and explain why it bleeds so much.

Set C · Tell it apart

1. The larynx is in the lower respiratory tract but in the conducting zone. Explain how both are true.
2. Distinguish edema from lymphedema by the structure at fault in each.
3. Compare a lymphatic capillary with a blood capillary by how each one begins and how permeable each one is.

DRAWING 3

The lymphatic system, and a node in section

BRAIN DUMP FIRST

Two minutes. Where the lymph goes, which duct drains what, and what happens inside a node.

Draw a body outline. Shade the upper right quadrant one way and the rest of the body another, then draw the right lymphatic duct draining the shaded quadrant and the thoracic duct draining everything else, both emptying where the subclavian and internal jugular veins meet. Add the cisterna chyli at the start of the thoracic duct and mark the cervical, axillary, and inguinal clusters. Beside the outline draw one node cut open, with its capsule, cortex, medulla, several afferent vessels, and one efferent vessel at the hilum.

In your notebook, head the page **M4-3 The lymphatic system, and a node in section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name which duct returns lymph from any region someone points to, and say why the two ducts are so unequal in size.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Two ways of grouping the parts

GRID

2 The three regions of the pharynx

LADDER

3 The larynx

HUB

4 The airway, trachea to alveolus

CHAIN

5 Right lung against left lung

SPLIT

6 The pathway of lymph, tissue space to bloodstream

CHAIN

7 A lymph node

HUB

8 Primary against secondary lymphatic organs

GRID